PORTFOLIO BUILDING B25CS0311

Activity 8: HackerRank Algorithmic Problem-Solving & Portfolio Integration

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GitHub: https://github.com/sanjaypindikoor9

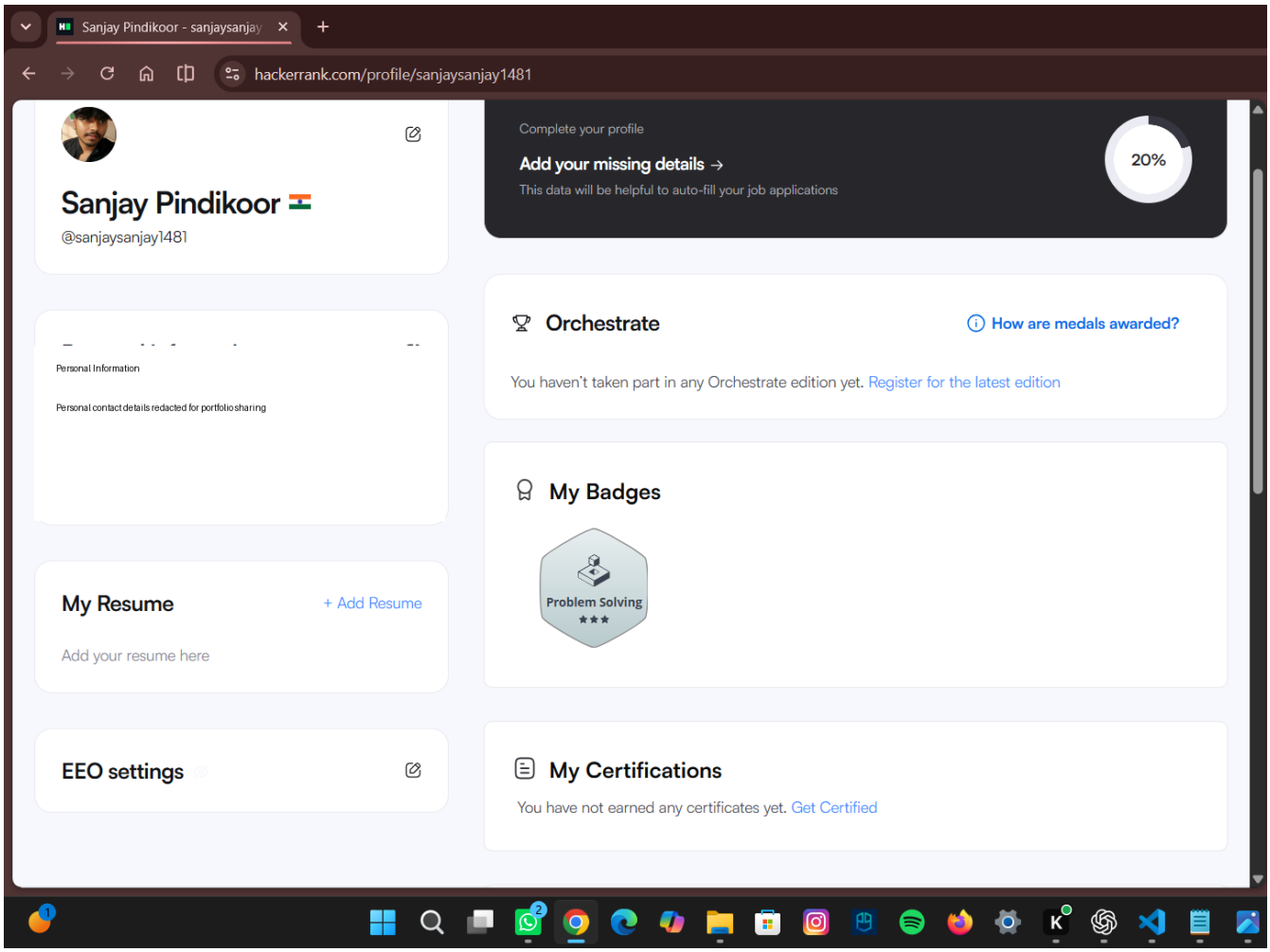
HackerRank: https://www.hackerrank.com/profile/sanjaysanjay1481

1. Problem-Solving Summary

Problem	Technique	Time	Space
Diagonal Difference	Single matrix traversal	O(N)	O(1)
Dynamic Array	Vectors + XOR index	O(N + Q)	O(N)
Time Conversion	Fixed-length string logic	O(1)	O(1)
Compare the Triplets	Element-wise comparison	O(1)	O(1)
Sparse Arrays	Frequency hash map	O(N + Q)	O(N)

2. HackerRank Evidence

The submitted profile screenshot verifies the public profile **@sanjaysanjay1481** and shows the required **Problem Solving 3-Star** badge. Accepted-submission screenshots for the five mandatory problems are included in the repository evidence folder.



3. Reflective Summary

This activity strengthened my understanding of selecting an appropriate data structure and keeping an algorithm efficient. In Diagonal Difference, I learned that a matrix problem can often be solved with one traversal by updating both diagonal sums at the same time. Dynamic Array demonstrated how vectors can represent a collection of variable-length sequences and how the XOR operation determines the sequence used by each query. Time Conversion showed that careful handling of boundary cases such as 12 AM and 12 PM is essential even when the algorithm is constant time. Compare the Triplets reinforced direct element-wise comparison and simple score tracking. Sparse Arrays highlighted the advantage of frequency mapping: instead of scanning the complete list for every query, a hash map stores counts once and answers each query with an expected constant-time lookup. Across the five problems, I learned to consider both theoretical complexity and implementation details such as auxiliary memory, input storage, and average hash-table performance. Organizing each solution with comments, a problem note, and a complexity analysis also showed me how algorithmic work can be documented as a professional GitHub portfolio. The next step is to keep practicing, improve complexity where possible, and maintain verified HackerRank evidence alongside the code.

4. Final Submission Checklist

✓ Five mandatory solution folders completed

✓ Root README with complexity table and repository structure

✓ Accepted-submission evidence for all five named problems

✓ Public HackerRank profile URL included

✓ Verified 3-Star Problem Solving badge screenshot included

✓ Repository ready to push publicly as HackerRank-3rdSem-Portfolio